

Section 3

PREPARING FOR THE FUTURE

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GENERATIVE AI AND THE TRANSFORMATION OF KNOWLEDGE WORK

by Maryam Alavi and George Westerman

Generative AI tools have taken the world by storm. ChatGPT reached 100 million monthly users faster than any internet application in history. The potential benefits of efficiency and productivity gains for knowledge-intensive firms are clear, and companies in industries such as professional services, health care, and finance are investing billions in adopting the technologies.

But the benefits for individual knowledge workers can be less clear. When technology can do many tasks that only humans could do in the past, what does it mean for knowledge workers? Generative AI can and will automate some of the tasks of knowledge workers, but that doesn't necessarily mean it will replace all of them. Generative AI can also help knowledge workers find more time to do meaningful work and improve performance and productivity. The difference is in how you use the tools.

In this article, we aim to explain how to do that well. First, to help employees and managers understand ways that generative AI can support knowledge work. And second, to identify steps that managers can take to help employees realize the potential benefits.

What Is Knowledge Work?

Knowledge work primarily involves cognitive processing of information to generate value-added outputs. It differs from manual labor in the materials used and the types of conversion processes involved. Knowledge work is typically dependent on advanced training and specialization in specific domains, gained over time through learning and experience. It includes both structured and unstructured tasks. Structured tasks are those with well-defined and well-understood inputs and outputs, as well as prespecified steps for converting inputs to outputs. Examples include payroll processing or scheduling meetings. Unstructured tasks are those where inputs, conversion procedures, or outputs are mostly ill-defined, underspecified, or unknown a priori. Examples include resolving interpersonal conflict, designing a product, or negotiating a salary.

Very few jobs are purely one or the other. Jobs consist of many tasks, some of which are structured and others which are unstructured. Some tasks are necessary but repetitive. Some are more creative or interesting. Some can be done alone, while others require working with other people. Some are common to everything the worker does, while others happen only for exceptions. As a knowledge worker, your job, then, is to manage this complex set of tasks to achieve their goals.

Computers have traditionally been good at performing structured tasks, but there are many tasks that only humans can do. Generative AI is changing the game, moving the boundaries of what computers can do and shrinking the sphere of tasks that remain purely human activities. While it can be worrisome to think about generative AI encroaching on knowledge work, we believe that the benefits can far outweigh the costs for most knowledge workers. But realizing the benefits requires taking action now to learn how to leverage generative AI in support of knowledge work.

How Can Generative AI Help?

When it comes to AI, it's often said that "AI won't replace you, but a person using AI will." Instead of automating your job away, the power of generative AI can help to improve your ability to do cognitively challenging knowledge work.

The key is to use generative AI to manage the flood of information washing past you every day. Humans have limited cognitive information processing capacity. On the other hand, most knowledge workers today are inundated with a high-velocity in-flow of digital information and always-on communications. This crush of information is creating a "digital debt": an ever-increasing backlog of information waiting to be processed by each knowledge worker.

If you feel this way, you are not alone. Sixty-eight percent of workers in a recent Microsoft survey indicated that they do not have sufficient uninterrupted time to focus on their core activities during the workday. The share of working hours taken up by emails, electronic meetings, text messages, and the search for and review of digital content is increasing even more.

This is where generative AI can help—if you use it wisely. In particular, generative AI can be useful in three major ways: reducing your cognitive load by automating some structured tasks, boosting your cognitive capacity for unstructured tasks, and improving the learning process for your job.

Reducing cognitive load

Generative AI tools can enhance performance and productivity by freeing mental capacity to focus on higher-value unstructured tasks. You can do this by offloading structured and repetitive elements of knowledge work to generative AI tools. In addition to reducing cognitive load, this can also make your job more interesting and satisfying by removing some of the drudgery involved.

Generative AI is already showing benefits in reducing cognitive load across a variety of industries. Attorneys at the global firm Allen & Overy use a system called Harvey to efficiently locate and access case law and draft simple contracts. This allows them more time to analyze complex legal issues and advise their clients.

In marketing and advertising, generative AI can automate routine content generation such as creating product brochures or personalizing email campaigns. A recent BCG survey of chief marketing officers found that two-thirds of respondents were investigating generative AI for personalization and half are exploring it for content generation.

In finance, a corporate bank is applying generative AI to reduce cognitive load from a constant inflow of new externally generated financial market information.¹ The system quickly analyzes and summarizes annual reports, earning call transcripts, and analyst reports to keep the bank's relationship managers

better informed about important developments. By streamlining the information search and review process, relationship managers have more time to focus and serve their clients.

As these cases demonstrate, delegating some of structured tasks to generative AI can help to relieve the stress of cognitive overload so you can focus on more important tasks. The drudgery can be done faster, and possibly better, by a computer, while you can improve your performance on the tasks that remain.

Boosting cognitive capabilities

Another approach to augmenting knowledge work is to use generative AI to boost higher-order cognitive processes to perform unstructured tasks. Three important areas are critical thinking, creativity, and knowledge sharing.

For critical thinking, generative AI can help people to ask better questions about the challenges they face. Experimenting in an executive education setting, researchers found that 94% of the time, engagement with AI (including generative AI) led to asking a wider range and variety of questions than the respondents otherwise would.² This in turn led to exploring ideas and possible solutions that they may not have considered, likely leading to better performance.

Another study found that ChatGPT was particularly useful in the idea generation and communication phases of strategy process.³ The AI tool created plausible strategic ideas with high efficiency. Its storytelling capability was particularly useful in helping to articulate and communicate the ideas. On the other hand, the tool was less helpful in suggesting ways to implement strategy, possibly due to the tool's lack of access to detailed information about the company, its capabilities, and other relevant contextual information.

University of Missouri professor Tojin T. Eapen and colleagues detailed how generative AI can promote divergent thinking by making connections among diverse concepts. Generative AI assisted not only in developing new ideas but also in evaluating and refining them based on criteria such as feasibility, impact, cost, and novelty.⁴ In a separate survey of more than 1,000 content creators (bloggers, podcasters, and short-form video producers), two-thirds of respondents indicated that they use the tools for creative tasks.⁵ Fifty-three percent said using the tools enhanced their creativity and productivity. Furthermore, those who had used the tools had a higher number of followers and had generated higher income.

Beyond generating knowledge, generative AI can also help to share it. Intellectual assets are dispersed across organizations in a wide variety of documents, policies, processes, and individual heads, making it difficult for people to access knowledge that already exists in the organization. By leveraging generative AI, companies can bridge the knowledge gap, facilitate knowledge sharing, and empower knowledge workers with the know-how they need to excel at their jobs. For example, to assist its wealth management advisers, Morgan Stanley implemented a generative AI model trained on a vast set of internally captured knowledge and expertise. Making wealth management knowledge readily accessible to every adviser in the company has been “transformative” by empowering advisers to efficiently address their clients’ specific questions and concerns.

Improving learning

Achieving mastery requires practice and not just classroom work. However, to be useful, practice requires feedback on performance—and providing personal feedback for every employee can be prohibitively expensive. The improving capabilities of generative AI are making it possible to consider the idea of an AI mentor for every knowledge worker.

For students, generative AI can play roles such as adviser, tutor, coach, and simulator. It can provide frequent feedback, personalized instructions and explanations, alternative viewpoints, and opportunities to practice through simulations. For example, Duolingo recently added two new generative AI–powered features to its language-learning software. Role-play allows users to practice free-flowing conversations in a foreign language, and Explain My Answer provides constructive personal feedback to users upon request.

Generative AI’s potential role in learning also extends to the work environment. Take, for example, the challenge of becoming a successful call center agent. Gaining skills in this complex, fast-paced environment requires a combination of instruction, practice, feedback, introspection, and immersion. At Fidelity Investments, trainees learn about a topic, then handle calls with a human mentor listening and advising, and then meet with the mentor and other trainees to discuss their experience. Generative AI can assist in this process, both during training and after. It can monitor a conversation with a customer and suggest what an agent can say or do to resolve the customer’s issues. It can also answer the agents’ questions later. A recent study of a call center introducing generative AI found that this type of support helped to improve productivity and quality for all workers while also increasing employee satisfaction.⁶ It also accelerated individual learning by enabling novice workers to progress more rapidly along the experience curve compared to those who did not leverage such a system.

How Managers Can Help Workers Use Generative AI

Hundreds of millions of people around the world are already using generative AI tools at home and at work. It’s likely that everyone is using generative AI differently, even if they’re using the same tools. This is an opportunity and a threat for companies. Each person’s experiments can generate innovations that could have great value. But some of the experiments may stray into dangerous territory by creating errors, breaking regulations, making biased decisions, or releasing private information to the public.

These tools are entering the sphere of knowledge work, whether wanted or not. Encouraging your people to experiment with the tools and try them in their jobs can help to alleviate some of their worries. This in turn can lead them to take a constructive and proactive stance toward working with AI, not fighting against it. To help your knowledge workers make the most of generative AI, consider these three key actions.

Define policies and assign responsibilities

While generative AI tools offer promising advantages, they can introduce risks to the organization. Therefore, it’s critical to understand and mitigate the risks of these tools and ensure that the procedures for safe use are followed. For example, using generative AI to write draft emails or summarize documents is easy and potentially highly productive. However, knowledge workers should be told not to do so with private information. Similarly, employees should know that while these tools are fast and easy, they are not always accurate. One does not want to find out about this weakness the way one lawyer did, when a judge cited him for including fictitious case law in his generative AI–drafted legal brief. Also, emphasize the biases that are created either in training or applying the tools and suggest ways to alleviate them.

Encourage experimentation and innovation sharing

Demonstrate the tools' capabilities in staff meetings and point to interesting use cases inside and outside of your company. As people share their innovations, ask probing questions to help everyone develop an awareness of the risks involved and know-how to mitigate those risks. Encourage peer learning, where employees teach each other how to use the tools. Building on that, ask workers to share the innovative practices they've developed, and take steps to help others adopt the best practices.

Celebrate the wins

By encouraging your team to treat generative AI as a voyage of discovery instead of a defensive activity, you can help them build confidence and capability in using the tool to enhance their cognitive ability and hone their job-specific skills. Make a big deal of the best case uses, and celebrate the innovations and the innovators. By doing so, you can not only improve the way workers use generative AI but also improve the innovative culture of your group.

Don't Wait

Knowledge work has always required ongoing learning to keep up with the progress of innovation and knowledge. Keeping up with the changes wrought by generative AI tools will require not only learning about the tools but using them to increase your knowledge and hopefully expand your role. In a recent McKinsey report, the nearly 1,700 executive respondents indicated that they expect more employees to be reskilled than laid off.⁷

The time to start working with generative AI is now. Generative AI can be a boon for knowledge work, but only if you use it in the right way. New generative AI-enabled tools are rapidly emerging to assist and transform knowledge work in industries ranging from education and finance to law and medicine. Companies are starting to introduce generative AI-powered innovations into their processes and to promulgate policies on how to use the tools safely.

However, there is no need to wait for these externally imposed changes. You can start now to use generative AI for your own benefit, once you understand and learn to mitigate the associated risks. Using free tools already available on the web, you can reduce your cognitive load from constantly rising tides of information while also boosting your cognitive abilities and learning effectiveness. Now is the time to start using generative AI in your knowledge work and to help your colleagues to use it wisely.



TAKEAWAYS

New generative AI-enabled tools, free and available on the web, are rapidly emerging to assist and transform knowledge work. Now is the time to start using generative AI—and to help your team and colleagues to use it wisely.

- # Generative AI tools can enhance performance and productivity by lowering cognitive load, freeing your mental capacity to focus on higher-value unstructured tasks.
- # Generative AI can help boost higher-order cognitive processes to perform unstructured tasks. Three important areas are critical thinking, creativity, and knowledge sharing.
- # Generative AI makes it possible to consider the idea of an AI mentor for every knowledge worker, which improves speed of learning.
- # Defining policies, assigning responsibilities, encouraging experimentation and innovation sharing, and celebrating wins is essential as you help your employees to explore and use generative AI.

NOTES

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WEB3 COULD CHANGE THE BUSINESS MODEL OF CREATIVE WORK

by Alex Tapscott

For most of the industrial age, technology has acted as a tailwind for creators, powering an age of mass culture that supported a professional class of working artists. However, more recently, the rise of *digital* technology—specifically, the internet and AI—has become a source of economic dislocation.

Web1, the so-called read web, was a digital printing press that democratized access to information, but it also commoditized art and music and undermined creators' rights as their IP got laundered in the web's swirling washing machine of content.

Web2, the read-write web, made it easier to publish content, share ideas and access a broader audience (i.e., "write" to the internet) but it also confined creators to tightly run platforms they did not trust or control and where they lacked transparency into the economic impact of their work. It was either that or risk the wilderness of the web, where their creations could be copied without recourse. Digital artists had no easy way to monetize their creations and could not benefit when works were resold.

The platforms that control content distribution have grown richer and more powerful, with precious little of that windfall trickling down to creators. Consider the strikes of the Writers Guild of America (WGA) and the Screen Actors Guild–American Federation of Television and Radio Artists (SAG–AFTRA). Streaming has already shortened television seasons, shrunk writers' rooms, and eroded artists' traditional residual revenues; and generative AI and 3D modeling give companies more tools for further minimizing their use of writers and actors.

By contrast, Web3—the read-write-own web—offers new tools to earn and own assets, build wealth, and wrestle back control from powerful platforms and intermediaries. Web3 could simplify how creators fund their ventures by crowdsourcing new titles directly from their fans. It could offer new ways to earn a living, not just on the first sale of a work of art but in perpetuity thanks to programmatic royalty streams paid via smart contracts, self-executing code that can move and store money. In other words, it might offer a new model for creative work.

AI: Risk or Opportunity?

Now, new technologies—namely, generative AI—pose new challenges for composers, screenwriters, visual artists, and other creative workers. It is too early to say whether AI will be all bad for all of them; it may expand the ranks of professional artists or grow the market for culture, creating opportunities for humans to cocreate or collaborate with AI on an equal footing. But it could also starkly devalue the work of copywriters, composers, and artists, and relegate these workers to supporting roles, such as polishing scripts or adding some depth of feeling to digitally rendered art. Cezanne said, “A work of art which did not begin in emotion is not art,” but plenty of studio executives see AI as a way to cut costs.

By analyzing hundreds of thousands of scripts against their viewership patterns over time, AI will get better at mimicking the stickiest and most binge-worthy of screenwriters’ styles, characters, and plots, not just to extend the arcs of existing stories but to synthesize all-new series and recommend the optimal cast or mix of character types. Ditto on the virtual replication of actors, with tools like FaceSwap already in use to de-age stars like Harrison Ford for his latest portrayal of Indiana Jones. But such big names likely consent to such innovations on their own terms. As *Vox* put it, “If people don’t lock down control of their digital twins, then nothing else matters.”

Oscar-winning director Steven Soderbergh told the *Hollywood Reporter* that, more than AI or visual effects, what keeps him up at night is the opacity of streaming data: He speculates that the studios are doing really well and don’t want to share the wealth or else really poorly and don’t want their stocks to tank.

Artists hoped that the internet would help to disintermediate gatekeepers and middlemen and change power dynamics in creative industries, but instead it added new intermediaries like streamers and platforms that distanced artists from their fans and obfuscated their economic impact. We need something different.

Culture Needs a New Business Model

If technology is upending the longstanding business model of creative industries, it’s also offering opportunities to establish a new one that can work better for creative workers.

Web3 technologies can simplify how creators track the usage of their IP and monetize it, ensuring they get paid promptly and fairly for their work. For instance, Web3 technologies could allow artists to grow and perhaps even thrive along with AI rather than suffer at its expense. For example, smart contracts can create avenues for artists to be compensated when their work is used to train an AI such as a large language model.

Web3 adds an *economic layer* and a *rights layer* to the internet stack, where users can not only track the provenance of information and intellectual property but also protect, manage, and monetize these digital assets themselves with transparency peer to peer. These innovations can also change how creative ventures are funded, removing industry gatekeepers and amplifying underrepresented voices. In the Philippines, independent game studios are selling non-fungible tokens (NFTs) of in-game assets to gamers directly to fund new titles, disintermediating big studios and traditional financial backers.

For Jules Urbach, CEO of cloud graphics company OTOY, Web3 is fit for this purpose. OTOY’s flagship product, OctaneRender, is what’s known as an “unbiased, spatially correct graphics processing unit (GPU) render engine,” which is the industry’s way of describing powerful software that can render more lifelike images and video than what came before. Marvel Studios used OctaneRender in the opening of *Ant-Man and the Wasp*, according to Urbach.

Using the cloud, OTOY is able to harness dozens—and sometimes hundreds—of GPUs at a time in its network to break down projects into smaller parts.

By breaking up tasks, OTOY helps democratize the compute-intensive process of rendering, allowing artists to render in a couple of minutes what used to take hours on expensive hardware setups. Most important, anything created in the render network creates a hash on a blockchain for the verifiable provenance essential to managing artists' IP rights and moving digital goods from platform to platform. With such capabilities, artist guilds could create a set of smart contract templates for their members to use in managing such lifelike renderings of themselves—their digital twins—and collect privatized data on industry usage that would give them bargaining power in future negotiations. The contracts could represent the artist guilds' negotiated terms, plus whatever the individual artists or their agents have negotiated.

From Hollywood to Everywhere

Web3 technology also offers artists ways to flip the traditional top-down model where studios and streamers try to control everything from IP to distribution. Hollywood screenwriter Jessie Nickson-Lopez has brought some of modern TV's most indelible characters to life. As a founding member of the writing team on *Stranger Things*, she developed the storyline for the character Eleven. Recently, as cofounder of Web3 startup MV3, she launched a collection of 6,500 NFTs of different characters, the building blocks for what will be a richly rendered narrative “universe,” originally created by Nickson-Lopez and her team. Set in the “dystopian cyberpunk society” of 2081, after the “climate has gone to hell,” MV3 focuses on “a ragtag group of idealistic” rebels struggling to “take power from the corporation that owns the city.” NFT owners will participate in the IP, have a say in their character's arc, and even cocreate the story with the MV3 team. These different character assets could ultimately appear in film, TV, and other storytelling media. They could be playable characters in video games or avatars in the metaverse.

Knowing that fans will support a project is key for firing up the Hollywood machine to spend \$100 million or more on a film or TV show. Nickson-Lopez said, “So we reverse-engineered it, and I created the world of Eluna City and the characters that reside in it.” Though Nickson-Lopez architected this world and crafted storylines for main characters, MV3 will not decide the direction of the story. “For me, what's been most exciting is seeing how much creativity we're inspiring in people who have never created before but are consumers of dystopian worlds and of fiction. There's this hunger to play. Our fans are our community and our cocreators. Because they're invested in the world and in the characters, they're really excited to build it up with us.”

MV3 inverts the Hollywood model. “As my lawyers say, ‘It's like you're ripping off pieces of value and just giving it to people.’ And we're like, ‘Exactly! That's exactly what we're doing.’ ” Recently, the founders handed day-to-day control of the MV3 universe to the community.

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Ultimately, projects like OctaneRender and MV3 will work because they engage their communities not only in the economic upside but also in the governance. Handing over control and economics to fans is antithetical to the Hollywood model, and Nickson-Lopez recognizes that. Engaging people in the creative process, for love of the experience and not just for money, is what clearly drives these creators and gives their projects purpose and meaning.

Technology tools and human capital are more distributed than ever. If Web1 and Web2 democratized access to information and made it easier to collaborate online, Web3 equips creators with a new tool kit to build real wealth from their work, on a globally level playing field. As the saying goes, the future is not to be predicted; it is to be achieved. For cultural industries and much else, Web3 can help us do it.



TAKEAWAYS

By offering new tools to earn and own assets, build wealth, and wrestle back control from powerful intermediaries, Web3 has the potential to simplify how creators—composers, screenwriters, visual artists, and others—fund their ventures and establish new ways to earn a living.

- # Web3's read-write-own technologies offer artists ways to flip the traditional top-down model where studios and streaming services aim to control everything from IP to distribution.
- # These innovations can also change how creative ventures are funded, removing industry gatekeepers and amplifying underrepresented voices.
- # Web3 makes it possible to incentivize creators to participate in creative and cultural communities not only for economic upside but also to be included in the governance of those communities.

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USING “DIGITAL ACADEMIES” TO CLOSE THE SKILLS GAP

by Rubén Mancha and Salvatore Parise

Today, the problem most companies face in executing their digital transformation is not access to technologies but a shortage of workers with digital and data science skills. The demand is growing faster than the supply. In other words, recruitment can only get your company so far. To get the rest of the way, companies need to commit to training and upskilling, giving employees opportunities to learn about technologies—and room to experiment (and fail) with them.

There are proven benefits to internal upskilling efforts, but also limitations. On the benefits side, organizations with more advanced upskilling programs report more significant innovation, digital transformation, productivity, business growth, and employee engagement and retention.¹ However, self-directed learning adds to the individuals’ workload—effort rarely acknowledged or rewarded by the organization. And widely available digital and data science learning platforms offer generic resources not linked to the organizations’ culture, innovation dynamics, and transformation opportunities, which will limit the value extracted from the workers’ gained skills.

Despite the benefits, only 18% of leaders “believe their organization has made ‘significant progress’ in establishing an upskilling program,” according to a survey of CEOs conducted by PwC.² Leaders reported struggling with delimiting the skills and allocating resources for upskilling initiatives, motivating employees, and retaining upskilled workers.

In our research on upskilling efforts, we found that “digital academies” are among the most successful approaches to closing the digital skills gap. A digital academy aims to catalyze how employees interact with digital and data science and lead the transformation of processes, products, and services. As a companywide effort, digital academies meet unique criteria:

- Digital academies are specific to the company’s culture and narrative.
- While diverse in length and content, their programs are highly experiential and considerate of organizational team dynamics.

- They reach across the enterprise to train new employees for technical career paths, knowledge workers on the use of specific technologies, and managers and executives to raise data science awareness and the impact of technologies in different areas of the business.

By focusing on how to use technology in the context of the organization and its digital vision, they help create and reinforce a specific culture around tech and innovation in a way that more generalized online trainings simply can't.

These have been built at companies such as DuPont, Gestamp, and Deloitte as a central initiative of their digital transformation efforts, resulting in increased capability of using data and adapting to technological advancements. In this article, we explain why digital academies work—and what companies need to keep in mind when designing their own.

DuPont's Spark Digital Academy

In 2020, DuPont, the multi-industrial specialty products company, implemented a digital transformation plan to become more innovative. To follow through on this ambition, however, the company “recognized that digital transformation was going to require the upskilling of our employees: both educate them on state of the art—what is possible—and change their skills,” said Duncan Coffey, DuPont's information and data science leader. There was a shortage of the necessary skills among current employees, and competition to recruit people with those skills was stiff.

In response, the company launched Spark Digital Academy in 2021. It aimed to create a bottom-up culture of data, agility, and experimentation across the organization; train workers on the use of data and digital technologies; and signal the company's priority in digital innovation. The course topics included technology (data science, robotic process automation, internet of things), capabilities (user experience, design thinking, agile/scrum, change management), and digital-specific product topics.

Since 2021, more than 500 DuPont employees have participated in instructor-led courses. The Spark Digital Academy has helped employees adopt digital technologies, optimize processes, and create value with digital products. The positive impacts of DuPont's upskilling program reinforce the foundation of the organization's digital transformation across all job roles and levels.

How can your company create a digital academy initiative? DuPont's efforts, in line with our research, suggest a few fundamental principles that lead to success.

Design Training That Will Serve Multiple Employee Segments

To facilitate the digital transformation of the organization, digital academies need to catalyze a digital and data science culture that works across the organization, helping practitioners, project leaders, and executives. When designing courses, it is important to connect to different target audiences in the organization and learn about their digital needs.

DuPont's digital academy served to train two primary roles: the “citizen data scientist” and the “translator.”

Citizen data scientists were employees who sought in-depth data science and digital expertise that they could apply directly to their job domains. They needed tools and software and access to professionally trained data scientists so they could build and implement models in their jobs. To address

their needs, the academy offered a foundational course for data science, followed by an advanced course for data science. Both courses typically required students to bring a project to do in parallel so they could apply the learnings immediately. Employees usually took them in series.

TABLE 11-1

Possible learning paths in data science for the “citizen data scientist” role		
Content	Modality	Duration
Curated tutorials on technologies and digital competencies	Self-paced materials, online	From a few hours to several days
Foundations of data science	Instructor-led course	Six hours per week, six weeks
Advanced data science	Instructor-led course with an experiential team project	Six hours per week, six weeks
Participation in communities of peer mentoring	On-demand	As needed

Translators, on the contrary, were employees who often had enough job experience and an end-to-end, big-picture view of work processes to understand how and where improvements could be made and wanted to bring digital solutions to science, engineering, and business problems. For them, digital and data science training opened a new lens for them to unlock value. The academy offered a data science course for senior leadership and subject business experts, providing general awareness of technologies and data science concepts.

Ensure Experiential Projects Are Part of the Digital Academy

Academies work best when employees bring on-the-job projects with them. This is true for a few reasons:

1. It ties digital principles to a real business problem, need, or opportunity so that essential concepts are not preserved in a vacuum or generic context.
2. Deeper learning occurs at work, not in the classroom.
3. It exposes students to less obvious problems that must be overcome on typical digital projects, such as quality data capture and cleaning.
4. Experiential projects in the classroom are also a way of derisking projects by quickly testing assumptions before spending large amounts of time, effort, and money.
5. Finally, it is a way to communicate value to business leaders so that they see the business value and can further sponsor and fund the effort.

For example, the DuPont digital academy’s latest courses on robotic process automation (RPA) included hands-on activities on cloud servers, so students could develop and implement RPA solutions during the course. Work was often done in groups to replicate team dynamics. The academy encouraged people from the same research group to sign up together for instructor-led courses with experiential components. One team project was started during the academy training. After communicating the project value to leadership, the team was then able to pull in extra resources to execute the project at a higher level when the course was completed.

Develop a Continuous Learning Relationship

One of the biggest mistakes we have seen with companies implementing digital academies is not continuing the relationship with employees after they have taken the course. More than simply helping employees acquire new digital skills, these programs create the space to initiate a process of cultural change. This process is fragile and needs time, frequent revitalization, and sustained access to resources. To ingrain a lasting digital mindset, academies need to develop an ongoing relationship with employee graduates. This should include:

- Providing services and support after each employee graduates through alum communities, mentor-mentee relationships, and technical assistance and infrastructure support, which is critical for digital and data science projects.
- Collecting continuous feedback from past graduates, internal business leaders, external vendors, and thought leaders on what topics should be offered or updated in the academy.
- Forging continued relationships—especially with project leaders and business leaders (or their direct reports) who took a course from the academy—is essential. These leaders often recommend employees or teams to enroll in a particular set of courses.

For example, DuPont's digital academy built communities of practice that could provide peer support on key topic areas, including a mentor-mentee program to support continued development work for those who completed the new RPA course. The academy also relied on input from leaders in DuPont's digital transformation groups—as well as from external instructors and facilitators—on where to grow the academy next. Finally, to raise awareness of the opportunities it offered, it implemented a badging and certificate program to build awareness, recognition, and advocacy. Course graduates received a digital badge that they could display on their email signatures and LinkedIn profiles.

Design the Digital Academy with Flexibility in Mind

In today's hybrid work environment with its increasingly uncertain work demands, companies should design their digital academies with course content and delivery flexibility to gain the highest potential demand. Academies that offer a portfolio of approaches—from self-paced on-demand courses to synchronous virtual or face-to-face courses—often achieve higher performance than those with a strict, narrow delivery format.

Academies should decide on a combination of experiential, peer, and formal instruction that works for the audience and the learning they seek. They should pay attention to self-paced content—not everything has to be instructor-led—and the value of experiential learning tied to the employees' roles. When possible, they should offer cases and content linked to the organization since this content is perceived as being more relevant. Also, employees with little or no experience with digital technologies may benefit from more direct instruction and guided activities. At the same time, advanced groups working on experiential projects may make the most of peer learning and open-ended project work.

Courses should also be designed with intensity and frequency in mind. For example, live courses that are spread out over days or weeks may be more accommodating and welcoming than courses that require several consecutive days of full-day participation. Popular instructor-led courses should also be repeated multiple times over the year.

At DuPont, their digital academy experimented with the delivery format of their data science foundation and advanced courses. Initially, they were delivered in an 8 a.m. to 5 p.m. format over a whole work week. After experimenting and receiving student feedback, a program of four hours per day over two weeks was very successful and better received.

Finally, providing learning pathways for their self-paced courses answered employees' "What course should I take next?" questions. The academy curated 20 self-paced tracks, which have generated thousands of views.

In conclusion, we should see the continued proliferation of corporate digital academies to upskill employees. This will have implications for online course providers such as Coursera and edX, consulting companies, and higher education institutions on how they deliver digital and data science training. For example, more colleges and universities offer specialized degree programs, certificates, and executive education focusing on this domain. Since many corporations interested in upskilling their workforce need more expertise, resources, and bandwidth to do digital academies completely themselves, we predict they will partner with these external institutions to design and deliver this training quickly and at scale.



Digital academies are among the most successful approaches to closing technological skill gaps. These internal upskilling programs can catalyze how employees interact with data science and lead the transformation of processes, products, and services. As companywide efforts, digital academies meet unique criteria and require new approaches.

- # They are specific to the company's culture and narrative. While diverse in length and content, their programs are highly experiential and considerate of organizational team dynamics.
- # They reach across the enterprise to train new employees for technical career paths, knowledge workers on the use of specific technologies, and managers and executives to raise data science awareness.
- # Many corporations interested in upskilling their workforce will be best served by partnering with external institutions to design and deliver this training quickly and at scale.

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NOTES

1. "Navigating the Rising Tide of Uncertainty," PwC, <https://www.pwc.com/gx/en/ceo-survey/2020/reports/pwc-23rd-global-ceo-survey.pdf>.
2. "Navigating the Rising Tide of Uncertainty."

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2

HOW TO PREPARE FOR A GEN AI FUTURE YOU CAN'T PREDICT

by Amy Webb

Recently, the CEO of a prominent bank phoned me to discuss the promise of generative AI. We initially worked through scenarios to improve fraud detection and customer service, but with the ongoing spate of recent announcements, it was clear he had grander ambitions in mind. Like many industries, banking has a workforce problem: There is a discrepancy between the demand for skilled personnel and the supply of workers who are willing to return to an office and play by pre-Covid-19 rules.

Generative AI, he thought, might be a silver bullet of sorts. It could create cost savings and efficiencies through automation, but might these new tools also solve the talent shortage issue? To put it plainly: How soon could AI replace human workers?

Our conversation echoed many I've had with executives across an array of businesses, including insurance, manufacturing, pharmaceuticals, and even executives leading Hollywood studios. They all want to know how their companies can create more value using fewer human resources. That's because when ChatGPT, the chatbot developed by OpenAI, suddenly went viral, it demonstrated the power of AI to generate its own emails, essays, recipes, financial reports, articles, and ideas. Goldman Sachs estimates that within the decade, 300 million jobs will either be eliminated or largely diminished by generative AI.

We're already starting to see turbulence. Job postings for "prompt engineers"—humans who ask systems like ChatGPT to generate content—are offering annual salaries of \$300,000 or more. OpenAI's GPT-4 passed the Uniform Bar Exam, hinting that, in the near future, we may not need lawyers for transactional work. Indeed, Walmart is prototyping a generative AI system (unrelated to OpenAI) to negotiate some of its vendor contracts; 75% of contract lawyers and procurement officers on the other side say they now prefer negotiating with an AI over their flesh-and-blood counterparts.¹ Google's Med-PaLM 2, which is a specialized model trained on medical knowledge, now answers medical exam problems at the expert level of a doctor. This summer, partners will start testing applications that can look at an X-ray and automatically write a mammography report—without a human doctor in the loop.

With the staggering pace of development, it's no wonder that so many executives are coming to the same conclusion: Within just a few years, powerful AI systems will perform cognitive work at the same level (or even above) their human workforce. Tempted by the possibilities of AI, concerned about finding and retaining qualified workers, and humbled by recent market corrections or missed analyst expectations, business leaders envision a future of work without nearly as many people as today. From my perspective, this is a huge miscalculation.

First, it's too early to predict the exact future of AI—especially given that generative AI is just one tiny area of a field with many interdependencies, each in various stages of development. Exactly which jobs AI will eliminate, and when, is guesswork. It isn't enough for an AI system to perform a task; the output has to be proven trustworthy, integrated into existing workstreams, and managed for compliance, risk, and regulatory issues.

Second, in a period of rapid disruption brought by technology, leaders are focused too narrowly on immediate gains rather than how their value network will transform in the future. As AI evolves, it will require entire segments of business to be reimagined—in real time but before we have a full sense of what the future will look like. Remember the earliest days of the public internet and web browsers, which were viewed as entertainment? No one planned for the fundamental transformation both would ignite. It would have been impossible then to foretell how it would someday influence presidential elections or create the world's first trillion-dollar companies.

To be sure, executives today must make decisions in the most complex operating environment I've seen since those early internet days. Leaders, understandably concerned about missing out on the next wave of technology, are unwittingly making risky bets on their companies' futures. Here are steps every leader should take to prepare for an uncertain world where generative AI and human workforces coexist but will evolve in ways that are unknowable.

Preparing for a Future You Can't Predict

Here's the paradox: We need to think of the workforce as evolving with—rather than being supplanted by—generative AI. The workforce will need to evolve, and workers will have to learn new skills, iteratively and over a period of years. Leaders must adopt a new approach to maximize the potential of AI in their organizations, which requires tracking key developments in AI differently, using an iterative process to cultivate a ready workforce, and most importantly, creating evidence-backed future scenarios that challenge conventional thinking within the organization.

What can leaders do now to navigate this period?

First, temper expectations about what generative AI can and will do for your business

Historically, AI cycles through phases that involve breakthroughs, surges of funding, and fleeting moments of mainstream interest, followed by missed expectations and funding clawbacks.

In 1970, Marvin Minsky, an influential computer scientist and one of the founding parents of AI, told *Life* magazine that artificial general intelligence—an AI with cognitive abilities indistinguishable from a person—was just three years away. Bear in mind that in the 1970s, the computing power required for such an AI didn't yet exist. Supercomputers were mostly theoretical. So were personal computers. The Datapoint 2200 and its processor eventually became the foundational architecture for what we came to know as PCs. The grand ambitions promised by Minsky and his colleagues never materialized, so funding and interest dried up. This happened again in 1987, when again, computer scientists and businesses made bold promises on a timeline for AI that was just never feasible.

While powerful, today's mainstream generative AI tools—ChatGPT, Midjourney, DALL-E 2—aren't finished products. Sometime soon, people will sour on their novelty and realize that while AI can create content, it's not good enough to actually use. Likewise, it's still very early days when it comes to domain-specific AI tools for medicine, climate, and life sciences. For generative AI to perform the miracles we've been promised—at scale and cost effectively—a lot more work needs to be done. Remember, these tools were mostly theoretical until very recently.

Executives need to get clear on the practical functions generative AI will perform in their organizations today. They should also be pragmatic about the opportunities—and risks—generative AI will eventually unlock. AI is not a monolith, and we are just at the beginning of a very long trajectory. This may sound intuitive, but in my observation, few leaders are developing a realistic strategy that links today's operations to tomorrow's vision, socializing it within their management teams, and revising their performance indicators accordingly.

Recently, I met with the executive leadership of a multinational consumer packaged goods (CPG) company eager to partner with a generative AI company. I walked them through a high-probability scenario in which customers using a chat tool answered a few questions about their preferences and goals and had an online shopping cart automatically filled with the items they would need for the week. But none of the CPG's brands showed up in the cart—or if they did, they weren't first on the list. Just as search engines like Google and Amazon invented new mechanisms and rules for search engine optimization, in the future, generative AI integrations across platforms like retailers and shopping cart apps would create new challenges for CPG companies, which might find themselves further down the value chain where critical decisions are made.

Second, evaluate what data your company is generating and how it would be used by generative AI, today and in the future

Business data is invaluable because once a model has been trained, it can be costly and technically cumbersome to port those data over to another system. At the moment, emerging AI platforms are not easily interoperable, and that's by design. Generative AI platforms are evolving into walled gardens, where the companies creating the technology control all sides of their ecosystems. The biggest AI companies are competing for market share—and for the enormous amounts of data they need to make their models most competitive. By marketing their platforms to companies, they want to lock them (and their data) in.

Today's AI systems are being created using a technique known as *reinforcement learning with human feedback*, or RLHF. Essentially, AI systems need constant human feedback or they run the risk of learning and remembering the wrong information. The more data that's ingested, the more annotating, labeling, and training that's required. Today, this work is automated to gig workers in emerging economies like Kenya and Pakistan. As AI matures, specialists with expert-level knowledge will be needed. Many of the business leaders I've met with aren't planning for a future that includes an internal RLHF unit tasked with continuously monitoring, auditing, and tweaking AI systems and tools. (The last thing any leader should want is an unsupervised AI system making decisions about how to improve itself.)

Even with trained humans in the loop, businesses must continuously craft scenarios that surface the risks of working alongside generative AI systems, especially those operated by third parties. That's because AI systems aren't static; they are improving incrementally over time. With each new development, new potential risks and opportunities arise. It would be impossible to game out all of the potential negative outcomes in advance without those predictions quickly becoming outdated. Instead, a

dedicated team should be charged with monitoring generative AI systems as they are learning, as well as related cybersecurity challenges, and they should develop short “what if” scenarios imagining ways in which things could go wrong.

Likewise, as AI evolves, so too will opportunities to unlock new growth. This means that businesses should also have a dedicated, internal business development team to develop near- and long-term scenarios for the myriad ways in which emerging tools will improve productivity and efficiency, lead to product development, spur innovation, and more.

Third, when it comes to AI, leaders must shift their focus from the bottom line to the top line

This will seem counterintuitive, as many view generative AI as a means to reduce operational costs. Today’s smart chatbots will soon give way to multimodal systems, which are AIs capable of solving different problems and accomplishing different goals at once. Imagine a property and casualty insurance company where every underwriter is teamed up with an AI. Initially, the underwriter might ask the AI to assess the risk associated with insuring a property; after a preliminary analysis of the text, she might ask it to refine results using the images from inspection reports or audio interviews with the prospective policyholder. She might go back and forth a few times, using different data sources, until an optimal quote is received for both the insurance company and the customer.

The key to making productive use of multimodal AIs is understanding how and what to delegate to a machine, so that both the human and the AI can accomplish more through collaboration than by working independently. However, delegation is something professionals routinely struggle with: They either assign too much or not enough, or not the right tasks. Working alongside a multimodal AI will require workers to master the art of delegation.

Once a workforce understands how to delegate correctly, it will act as a force multiplier within organizations. Individual teams could be more ambitious in growing the company’s top line through ideating and simulating new revenue streams, finding and acquiring new customers, and seeking out various improvements to the company’s overall operations.

This portends a future that demands a different approach to upskilling. Most workers won’t need to learn how to code, or how to write basic prompts, as we often hear at conferences. Rather, they’ll need to learn how to leverage multimodal AI to do more, and better, work. Just look at Excel, which is used by 750 million knowledge workers every day. The software includes more than 500 functions—but the vast majority of people only use a few dozen, because they don’t fully understand how to match the enormous number of features Excel offers to their daily cognitive tasks. Now, imagine a future in which AI—a far more complicated, more convoluted software—is ubiquitous. How much utility will be left on the table simply because business leaders approached upskilling too narrowly?

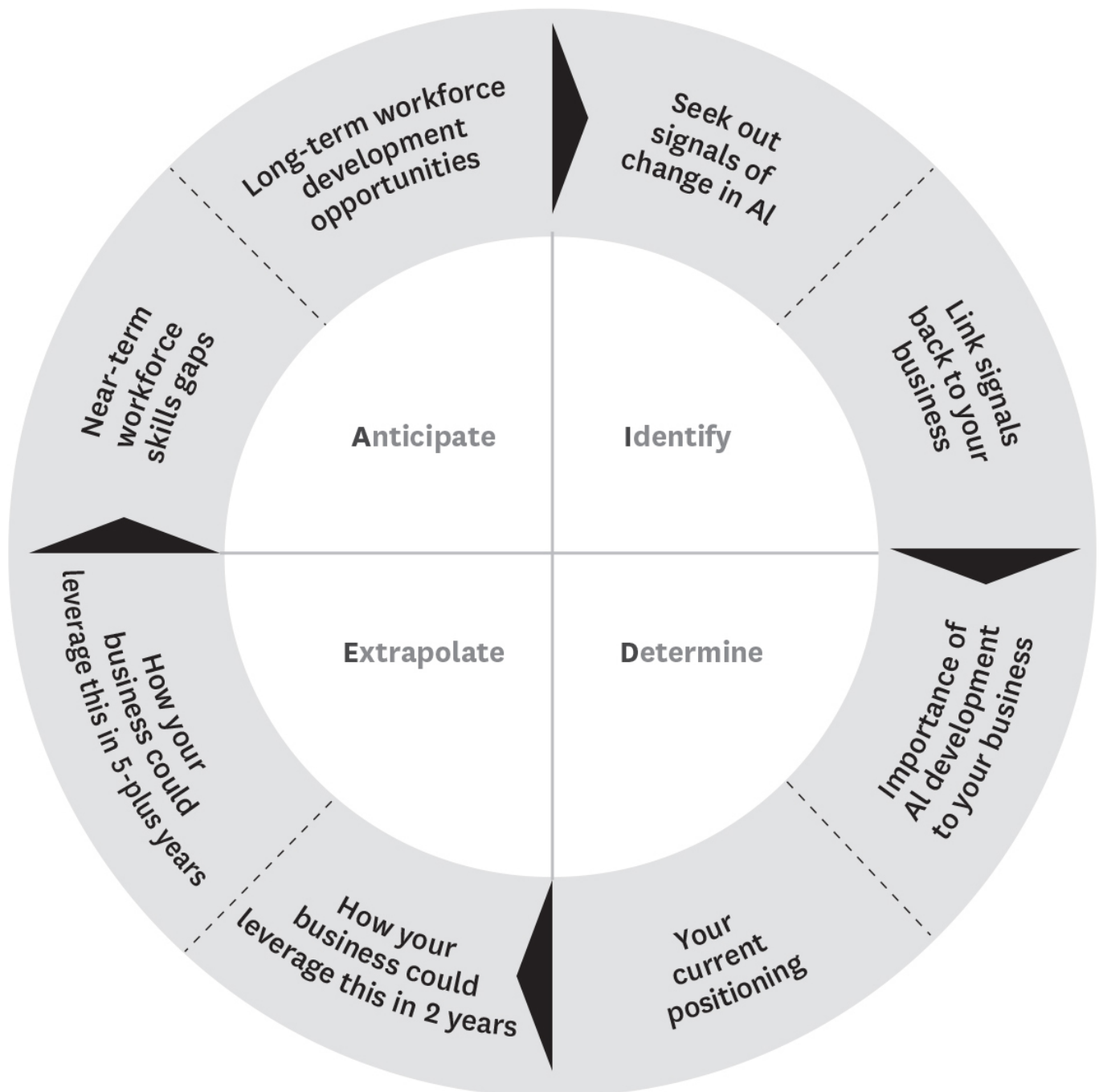
A Framework for Navigating the Evolving AI Workforce

Workforce change is an inevitable side effect of technological evolution, and leaders need a systemized way of seeing what the future of their organizations will look like in the wake of generative AI’s developments. To that end, the simple framework in [figure 12-1](#) will help leaders in any organization anticipate how—and when—their workforce will need to change in order to leverage AI. The goal isn’t to make long-range predictions, or even to be ready for everything—it’s to position organizations to be ready for anything as AI continues to improve.

FIGURE 12-1

The IDEA framework

This four-step approach helps predict the dynamics of how workforces will transform. Following the steps of **identify**, **determine**, **extrapolate**, and **anticipate** can help leaders see risk and opportunity early enough for action. Used regularly, this framework enables you to see the landscape more clearly, evaluate gaps within your organization, and link emerging technology to existing strategy, positioning you to make decisions with confidence.



This framework should be used to develop scenarios for the future of a business. It is designed to help you see risk and opportunity early enough for action. Used regularly, this framework enables leaders to see the landscape more clearly, evaluate gaps within their organizations, and link emerging technology to existing strategy, positioning them to make decisions with confidence. Importantly, it asks leaders to

think exponentially about AI but to act incrementally in response to new developments. While it won't predict a singular future for your company—no scenario can do that—it will prepare leaders to make decisions well ahead of their competitors.

The single best thing organizations can do right now—during this period of what feels like a soul-crushing amount of change and uncertainty—is to methodically plan for the future. That requires knowing generative AI's limitations as well as its strengths and adopting a culture of continual evaluation and improvement. It also means getting past clever product demos to much more mundane, pragmatic conversations about the trajectory of development, how data are being used, and the practical ways in which companies can use emerging tools. Resist the temptation to reduce your workforce—and instead use strategic foresight to create a future where AI is leveraged by a highly skilled workforce, and where human-AI teams are more productive, creative, and efficient working together than apart.



TAKEAWAYS

Given the staggering pace of development in generative AI, organizations must be methodically planning for the future. That requires knowing generative AI's limitations as well as its strengths and adopting a culture of continuous evaluation and improvement. Every leader should take the following steps to prepare:

- # **Temper expectations.** Historically, AI cycles through phases that involve breakthroughs, surges of funding, and fleeting moments of mainstream interest, followed by missed expectations and funding clawbacks.
- # **Evaluate how your company's data could be used by generative AI today and in the future.** Businesses should have a dedicated internal business development team to develop short- and long-term scenarios.
- # **Stop focusing on lowering operational costs; start focusing on creating new business models.** Learn how and what to delegate to machines, so that both the human and the AI can accomplish more through collaboration than by working independently.

NOTES

1. "Walmart Reportedly Finds 75% of Vendors Prefer Negotiating with Chatbot," PYMNTS.com, April 26, 2023, <https://www.pymnts.com/news/artificial-intelligence/2023/walmart-finds-75-percent-vendors-prefer-negotiating-with-chatbot/>.

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